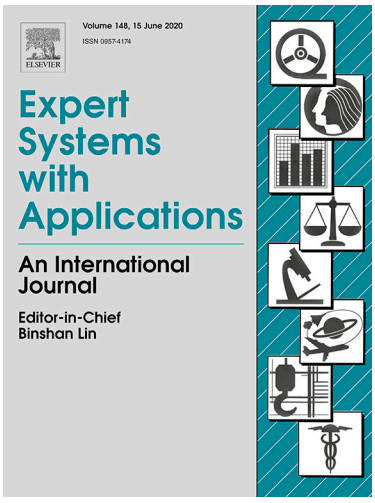
Journal Pre-proofs



Corporate Default Forecasting with Machine Learning

Mirko Moscatelli, Fabio Parlapiano, Simone Narizzano, Gianluca Viggiano

PII: S0957-4174(20)30391-2
DOI: https://doi.org/10.1016/j.eswa.2020.113567
Reference: ESWA 113567

To appear in: *Expert Systems with Applications*

Received Date: 24 July 2019
Revised Date: 12 March 2020
Accepted Date: 13 May 2020

Please cite this article as: Moscatelli, M., Parlapiano, F., Narizzano, S., Viggiano, G., Corporate Default Forecasting with Machine Learning, *Expert Systems with Applications* (2020), doi: https://doi.org/10.1016/j.eswa.2020.113567

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2020 Published by Elsevier Ltd.

Corporate Default Forecasting with Machine Learning

Mirko Moscatelli

Research, Economics and Statistics Department, Bank of Italy, Via Nazionale 91 - 00181- Rome (Italy)

mirko.moscatelli@bancaditalia.it

Fabio Parlapiano

Research, Economics and Statistics Department, Bank of Italy, Via Nazionale 91 - 00181- Rome (Italy)

fabio.parlapiano@bancaditalia.it

Simone Narizzano

Markets and Payments Systems Department, Bank of Italy, Via Nazionale 91 - 00181- Rome (Italy)

simone.narizzano@bancaditalia.it

Gianluca Viggiano

Markets and Payments Systems Department, Bank of Italy, Via Nazionale 91 - 00181- Rome (Italy)

gianluca.viggiano@bancaditalia.it

(1)

June 2019

Abstract

We analyze the performance of a set of machine learning models in predicting default risk, using standard statistical models, such as the logistic regression, as a benchmark. When only a limited information set is available, for example in the case of an external assessment of credit risk, we find that machine learning models provide substantial gains in discriminatory power and precision, relative to statistical models. This advantage diminishes when confidential information, such as credit behavioral indicators, is also available, and it becomes negligible when the dataset is small. Moreover, we evaluate the consequences of using a credit allocation rule based on machine learning ratings on the overall supply of credit and the number of borrowers gaining access to credit. Machine learning models concentrate a greater extent of credit towards safer and larger borrowers, which would result in lower credit losses for their lenders.

Keywords: Credit Scoring, Machine Learning, Random Forest, Gradient Boosting Machine

JEL Codes: G2, C52, C55, D83

[1] The authors wish to thank Giorgio Gobbi and Francesco Columba for their comments.

Journal Pre-proofs

1. INTRODUCTION

Default forecasting is of key importance for financial institutions and investors. For instance, banks use probabilities of default (PD) to screen potential borrowers, evaluate the terms of new loans and manage the risks stemming from lending activities. Investors also make extensive use of PD and probabilities of migration across different credit rating classes both for bond pricing and portfolio management. In addition, macroprudential authorities have an interest in the surveillance of default risk as it represents a major source of risk for lenders.

Since the Basel II accord, default forecasting methods based on a reduced-form regression approach have become popular in the banking industry. These methods consist of multivariate regression models which use firms' characteristics, such as economic and financial fundamentals, to predict the credit quality of a firm. Their outcome is a credit score, a continuous numerical indicator of creditworthiness which translates into a probability of default. Although the models are calibrated with statistical techniques, the selection of relevant predictors reflects *a priori* choices based on economic reasoning about the structural characteristics that explain the state of health of a company; moreover, they provide insight about the significance of individual characteristics for firms' default risk (Sironi and Resti, 2007). Overall, statistical models provide a satisfactory forecasting device, accommodating both accuracy and transparency requirements. This quality is due to their plain functional form, which combines additively monotonic predictors of default into a probability with good out-of-sample performance.

The Global Financial Crisis, however, highlighted some pitfalls in default forecasting based on commonly used rating systems approaches: i) their slow capacity to adapt to changes in the state of the economy, and ii) their limited ability to model complex non-linear interactions between economic, financial and credit variables. For example, credit ratings may overlook signals of a deteriorating economic and credit environment, such as a rapid increase in default rates or negative shocks to the supply of credit.

Availability of large datasets and unstructured information also pose a challenge to parsimonious rating systems based on a regression approach, since their predefined functional form may inhibit full exploitation of available information. Additionally, from a macroeconomic perspective, not all defaults entail equal consequences for the economy. For example, defaults or early stages of credit deterioration of Sovereign borrowers are known to have a ripple effect through the economy. In a similar fashion, the default of large corporate borrowers may spill-over to other connected entities (partners, suppliers or customers) and impair their creditworthiness. Thus, being able to detect these infrequent, but critical default events, is a desirable property of a default forecasting model.

A growing strand of research suggests that the use of models based on Machine Learning algorithms (ML) to predict defaults provides a suitable alternative to the modelling of default risk. ML define a class of models that can perform complex forecasting tasks when the relationship between predictors and the outcome is unclear or unknown. Models based on ML (henceforth ML models) typically neglect the inference about the importance of economic determinants for default risk but, in turn, highly accurate out-of-sample forecasts can be obtained with ML as established in a number of works (Baesens at al., 2003; Brown and Mues, 2012; Barboza at al., 2017). This feature of ML models is particularly relevant to credit risk applications, but it comes at the cost of lower transparency relative to statistical models: ML models do not provide estimates of the parameters that relate predictors to the outcome variable (the models are non-parametric), and such "black box" feature can make their rationale and their forecasts difficult to explain.

This work aims to investigate how default forecasting can benefit from the use of ML algorithms. We contrast statistical models with ensemble decision trees, a class of ML models which can handle both complex relationships across different variables (e.g. non-linear or non-additive) and large datasets with correlated predictors. A decision tree algorithm is in essence a collection of rules that recursively partition the full set of firms into homogeneous sub-sets according to their characteristics and the outcome variable (i.e. default and non-default). Predictions are then obtained as the odds of a given outcome in each subset. In this work we use random forest (RDF) and gradient boosted trees (GBT) models, which combine a large number of predictions stemming from individual decision trees into a single (ensemble) highly accurate forecast; this method mitigates the tendency of decision trees to overfit the data. The two models differ in the way individual trees are grown. In the RDF model, a random sample of the data and a random selection of variables are used for each tree, so as to obtain less correlated individual predictions. The GBT model, instead, combines predictions obtained from trees that sequentially specialize in dealing with the forecasting errors of their predecessors.

We estimate the models using a large dataset covering financial and credit behavioral indicators for Italian non-financial firms. We test the out-of-sample performance for these competing models comparing one-year-ahead PD estimates and observed default data for the 2011-17 period.

The analysis highlights the following main results:

(i) when only publicly available information is used to train the models, ML outperforms statistical models both in terms of discriminatory power (the capacity to rank borrowers according to their riskiness) and precision (the ability to estimate PDs that deviate only marginally from actual default rates). The gains from ML remain, but to a lesser extent, when the quality of the training dataset is enriched with high quality confidential information. Finally, when the number of observations is

small, as in the case of datasets available to lenders, the gains in forecasting performance from the use of ML are negligible;

(ii) a credit allocation rule based on borrowers' default probabilities results in a larger supply of credit and, at the same time, a lower default rate when using PDs obtained with ML;

(iii) with ML rating systems, safer borrowers - which gain prioritized access to credit - are usually larger borrowers, and this results in a more contained pool of borrowers for a given amount of available credit;

(iv) we argue that improvements in forecasting performance from the use of ML are due to its capacity to exploit complex relationships between predictors and default outcomes: indicators presenting a non-linear relationship with the default outcome are more important for ML than for statistical models.

The use of the ML approach in credit risk modelling has gained momentum with recent applications in the field of early warning systems for banking crises, predictions of household mortgage or consumer credit default, and corporate default.

Chakraborty and Joseph (2017) train a set of models for the prediction of financial distress based on balance sheet items of financial institutions, finding that ML approaches generally outperform statistical models such as the logistic regression. RDF, in particular, provide a marked increase in discriminatory power of about 10 percentage points in terms of Area Under the Receiver Operating Characteristic (AuROC) compared to the logit model.

Using data on U.S. households' defaults on mortgages, Fuster et al. (2018) find that the RDF model generates more accurate predictions than those of the logit model, although the improvement is small and accounts for about 1.2 percentage points of AuROC. Most of these gains result from the sophisticated functional form of the RDF model, which captures the complex relationships connecting different variables to default outcomes with greater discriminatory power.

The modelling of consumers' credit risk is by far the most popular application for ML algorithms in default forecasting. Khandani et al. (2010) apply ML and statistical models to predict consumers' defaults using a rich set of customer transactions and credit bureau data; their work shows that RDF forecasts attain an improvement over standard statistical techniques for measuring credit quality, and their use would yield a substantial reduction of losses, from 6 to 23 percent of banks' total assets. Other applications to consumer credit risk also confirm the robustness of RDF predictions, see for example Kruppa et al., (2013); Yuan, (2015).

The applications of ML to corporate default forecasting are also numerous. Using a large dataset covering the North American corporate sector for the period 1987-2013, Barboza et al. (2017) show that ML provides a substantial improvement by about 10 percentage points of AuROC in relation to traditional models. In Bachman and Zhao (2017), the performance of ML models are compared to the Moody's proprietary algorithm based on a regression model using corporate data for the US; the exercise shows that ML models deliver AuROC of about 2-3 percentage points higher than those of the regression approach. However, their opaque structure may lead to PDs that are difficult to relate to underlying firms' characteristics. Furthermore, the inclusion of credit behavioral variables within the set of predictors notably increases the discriminatory power of each model by over 10 percentage points (in terms of AuROC). Fantazzini and Figini (2009) compare RDF and logit model forecasts of the probability of survival using SMEs data in Germany for the 1996-2004 period. Their work documents a weak association between in-sample and out-of-sample forecasting performance, highlighting overfitting problems that can be associated to ML. In contrast to previous evidence for default probabilities, firms' survivals out-of-sample forecasts obtained using the logit model outperform their RDF counterparts.

Our results contribute to the literature on corporate default forecasting in a number of ways. First, we document to which extent there is a gain in accuracy when using ML vis-à-vis statistical models. The comparison covers different sets of data and different groups of borrowers. Second, differently from previous studies, we document the precision of PDs estimates using backtesting; these results complement the existing evidence about the discriminatory power of ML models and provide practical implications for potential users of ML algorithms. Third, we study the impact of using ML for credit allocation, analyzing the changes in the overall supply of credit, the number of borrowers gaining access to credit and the default rate.

With the aim of promoting fair lending practices, the US banking regulation has recently introduced the obligation for creditors to provide notice of the adverse factors underlying a credit application.[2] This provision exacerbates the trade-off between using transparent statistical models and adopting more accurate but difficult to explain models. Nevertheless, our analysis sheds light on the circumstances for which the more opaque ML approach provides superior forecasts, warning lenders to cross-reference their PDs obtained under statistical models with those obtained under ML models.

[2] The Equal Credit Opportunity Act (ECOA) and the Fair Credit Reporting Act (FCRA) provide lenders with the obligation to inform the borrower about the factors involved in taking actions that are adverse or unfavorable to the positive outcome of a credit application. In turn, these requirements provide actionable information to improve borrowers' credit standing. See What Are We Learning about Artificial Intelligence in Financial Services? speech from the Governor Lael Brainard at "Fintech and the New Financial Landscape", November 13, 2018, Philadelphia, Pennsylvania.

The rest of the paper is organized as follows. Section 2 reviews the credit risk models employed to estimate probabilities of default; section 3 describes the dataset and the estimation process; section 4 presents the results; section 5 concludes.

2. CREDIT RISK MODELS AT A GLANCE

We compare two different classes of default forecasting models: statistical and ML models. Some key differences characterize the two. On the one hand, statistical models are particularly fit for the purpose of inference, and typically rely on many assumptions regarding the structural relationships between variables, the number of parameters that can be robustly estimated, and the distribution properties of the data generating process. On the other hand, ML models are mostly focused on prediction accuracy, and make very weak assumptions on the structure of the data generating process. This feature allows the detection of data-driven interactions and non-linear or non-monotonous relationships between predictors and the outcome variable. Moreover, ML often involve the estimation of multiple models rather than a single model (this is the main reason why ML are more computationally intensive), and use only the most accurate model to perform prediction tasks. In this section we briefly review standard credit risk models (2.1) and introduce ML models that are used in the empirical application, namely random forest and gradient boosted trees (2.2).

2.1 STATISTICAL MODELS

Statistical theory offers a variety of methods for the estimation of default probabilities; linear discriminant analysis and logistic regression are the most popular. The linear discriminant analysis (LDA) provides an assessment of corporates' credit quality using a linear discriminant function that classifies corporate borrowers into groups (default and non-default) based on their characteristics. For more than three decades the discriminant analysis recorded extensive use by practitioners, with a reasonably good performance in predicting bankruptcy and other types of distress of both privately and publicly held non-financial firms in the international context (Altman's, 1968; Altman, 1983; Altman et al., 2017). However criticisms regarding its underlying assumptions (firms belonging to two different populations, the normal distribution of observables, and equal covariance matrices for the two populations) gradually opened this field to more flexible models such as the logistic regression.

The Logistic regression model (LOG) provides estimates of a continuous probability of default from firms' observable characteristics using the two extreme values for the probability of default itself as the dependent variable: 0 for financially sound firms and 1 for defaulted firms. The model assumes that firms belong to the same population and that a known structural relationship (additive and linear) exists between the observable characteristics of the firm and the probability of default. The parameters of the model are usually estimated

using maximum likelihood. The LOG relaxes some of the assumptions needed for discriminant analysis (multivariate normality and equal covariance matrices), and it has the important advantage of ease of interpretation of its results, delivering a straightforward probability of default for individual borrowers.

These features of the LOG attracted a strong interest and prompted its diffusion as a scoring model amongst practitioners. Nevertheless, a number of limits still apply: the lack of consideration of non-linear or complex interactions between observables and defaults, the sensitivity to outliers or missing data, and the difficulty to fully exploit large datasets. The Penalized Logistic Regression model (PLR), also known as the elastic net regularization of the logistic regression, combines the linear nature of the previous models with the focus on prediction accuracy typical of ML models. The PLR has the same structural form of the logistic regression, but it estimates the parameters using an objective function aimed at maximizing out-of-sample forecasting performance.

Previous evidence shows that PLR can outperform the standard logistic regression in some prediction tasks (Zou and Hastie, 2005); however, the limitations of parametric models still apply.

2.2 Machine Learning models

We use two powerful ML algorithms which found extensive use in credit risk applications: Random Forest (Breiman, 2001) and Gradient Boosted Trees (Friedman, 2000). The building blocks for these models are classification trees, partition algorithms that recursively splits the dataset into smaller sets (or branches) that best separate defaulters from non-defaulters.[3] At each iteration, the decision tree algorithm chooses from the covariates space $\mathbf{X}$ a variable and a value for that variable, so as to minimize a measure of heterogeneity (impurity index) in the resulting subgroups with respect to the classification variable.[4] The process continues within each branch until a stopping condition is reached, such as too few observations in the branch or no significant reduction in impurity. The final branches - called leaves - contain a constant estimate for the probability of default of firms in each leaf, computed as the proportion of defaulted over total number of firms in that leaf in the estimation (or training) dataset.

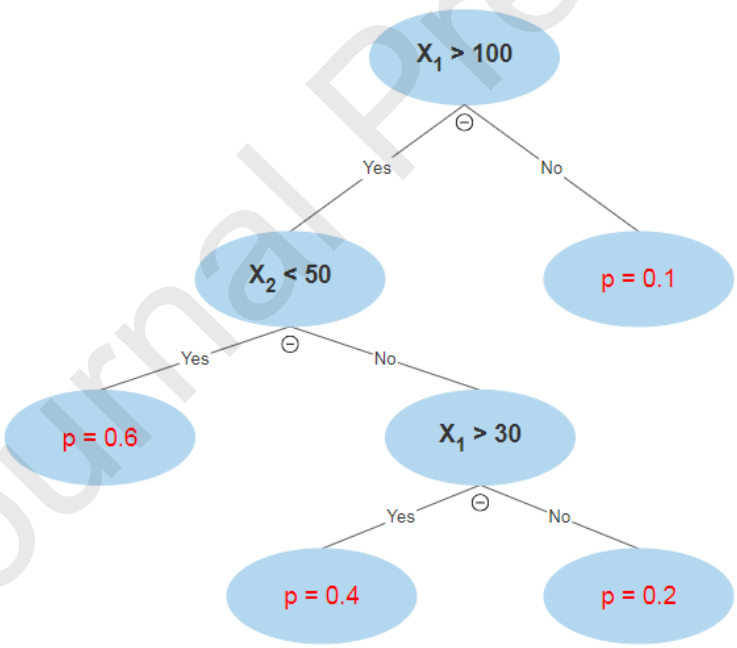
[3] A good introduction to trees, random forest and boosting algorithms can be found in Hastie, Tibshirani and Friedman (2001).

[4] An example of measure of impurity, as well as the one that we use, is the Gini coefficient. Given a set of observations S with a binary value for each observation $y_i \in \{0,1\}$, the Gini impurity coefficient is defined as $GI(S) := 2 \cdot \sum_i p_i(1 - p_i)$, where p_i is the percentage of observations in S such that $y_i = 1$. The Gini Impurity ranges between 0, when $p_i = 0$ or $p_i = 1$, and 0.5 when $p_i = 0.5$.

A simplified classification tree containing only two splitting variables X_1 and X_2 and four leaves is shown in Figure 1. The first level of the tree splits the whole sample into two branches depending on the value of X_1. Firms with X_1 less or equal than 100 end up in a leaf and receive a default probability equal to 0.1, whereas firms with X_1 greater than 100 are classified in a middle branch and are divided further according to the value of X_2. Those firms with X_2 smaller than 50 end up in a leaf and receive a default probability equal to 0.6, while firms with X_2 greater or equal than 50 undergo a final split according to the value of X_1: if X_1 is greater than 30 their default probability equals 0.4; otherwise, they receive a default probability of 0.2.

Notice that the classification tree model: i) captures interactions between variables: the effect of X_2 on the default probability strongly depends on the value of X_1,for instance, if X_1 is less or equal than 100 the value of X_2 is irrelevant for the estimated default probability; ii) captures non-linear relationships, since the same variable can be used more than once with different values to split the tree, for example, X_1 is used twice to split the sample; iii) the default probabilities are not a continuous function of the variables: a small increase from 100 to 101 in the value of X_1 results in the default probability of the firm going from 0.1 to 0.2 at least.

Figure 1: Decision Tree



Note: A decision tree for predicting firms' default probabilities grown using two variables, namely X_1 and X_2. Decision rules according to which branches are split are reported in black, while estimated default probabilities are reported in red.

Classification trees have a desirable low-bias property, meaning that the leaves - which are defined using multiple variables simultaneously - can fit the data extremely well. This flexibility, however, results in an

undesirable high-variance property of the model, meaning that out-of-sample predictions are highly sensitive to small changes in the underlying estimation dataset; this often leads to a low forecasting power of the model.

This shortcoming is addressed by ensemble classifiers, such as those provided by the Random Forest and Gradient Boosted Trees models. Instead of using a single classification tree, these models grow a large set of trees, and the final prediction is obtained as the average of the predictions stemming from the individual trees. In particular, the predictive function F takes the form of $\mathrm{F(x)} = \frac{1}{\mathrm{k}}\sum_{\mathrm{i}=1}^{\mathrm{k}}\mathrm{T_i(x)}$, where $\{\mathrm{T_1}, \ldots, \mathrm{T_k}\}$ are the k different classification trees and $\mathrm{T_i(x)}$ is the probability of default that the tree $\mathrm{T_i}$ associates to a borrower with x covariates. The two models differ according to how the different set of trees is constructed.

The Random Forest model grows the set of trees i) using for each tree a different bootstrapped sample of the original dataset (i.e. a sample with replacement having the same number of observations of the original dataset), and ii) selecting at each branch the best split using only a randomly selected subset of the covariates. This procedure implies that the trees differ from one other, since the underlying information set is different.

The Gradient Boosted Trees model grows the set of trees recursively, using a learning from mistakes approach, where at each step classification errors from the previous trees are used as the dependent variable to grow the next tree. The process proceeds as follows: the first tree T_1 is a standard classification tree as described above; from the second tree onwards, trees are grown using the same x covariate set but a different dependent variable, computed as the difference between the $0/1$ default outcome and the estimations of the previous trees.[5] In other words, the first tree T_1 will be trained on the model $y = T_1(x)$, the second tree T_2 will be trained on the model $y - T_1(x) = T_2(x)$, the third tree T_3 on the model $y - T_1(x) - T_2(x) = T_3(x)$, and so on, each time trying to predict the forecasting errors of the previous trees.[6] This procedure of continuously learning from previous forecasting errors can achieve very accurate predictions but can also lead to overfitting, thus the number of trees is a very important feature of the model (hyper parameter) that is chosen using cross-validation.[7]

3. The training dataset

3.1 Corporate defaults

[5] Regression trees differ from classification trees because the output variables is continuous rather than numerical. They are generated in the same way, with the only difference that the impurity function is the variance of the outcome variable instead of the Gini impurity.

[6] Residuals can be interpreted as negative gradients in F of the quadratic loss function $\frac{1}{2}(y - F(x))^2$ from which derives the name "gradient boosting".

[7] Hyperparameters define the general characteristics of a model, such as its complexity, and can either be set apriori or learned from the data through optimizers such as grid search.

We use an extensive dataset of financial and credit behavioral indicators for Italian non-financial firms for the years 2011-17. Our dependent variable, the financial default, is sourced from the Italian Credit Register (CR) and reflects a system-wide definition of non-performing status of a borrower. A firm is classified in default in a given year if the ratio of non-performing credit to total credit drawn from the banking system is greater than 5 percent for at least one month.[8] The default rate, i.e. the ratio of borrowers classified as non-performing in a given year to the total number of borrowers not in default at the beginning of the year, gives an aggregate measure of credit risk which we aim to model at the firm level (Table 1).

Within the 2011-17 time period, credit risk stemming from the corporate sector peaked in 2014, in the aftermath of the European Sovereign Debt crisis and of the associated slowdown of the Italian economy. Following the monetary measures undertaken by the European Central Bank, the gradual improvement in the business cycle and the exit of vulnerable firms from the market, the aggregate default risk also decreased, with default rate levels approaching about 2.5 percent in 2017.

Table 1: Default rate

Year	N Firms	Default Rate
2011	222.879	1,20%
2012	233.157	2,45%
2013	259.166	2,23%
2014	249.566	4,76%
2015	252.059	4,12%
2016	260.156	3,31%
2017	269.657	2,53%

Note: Our calculation on the Credit Register data. N Firms reports the number of firms included in our sample, while Default rate is the proportion of firms in default in a given year over the number of firm not in default at the beginning of the year.

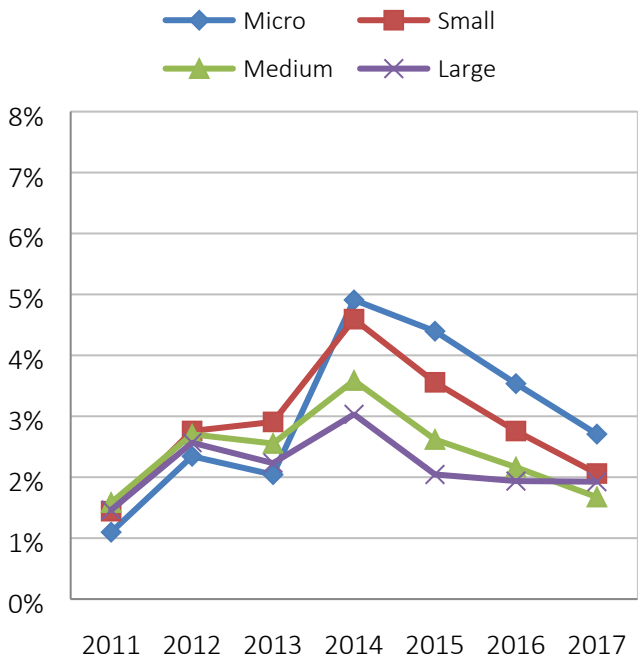
The sudden increase in the number of defaults both in 2012 and 2014, that more than doubled compared to the previous year, may pose a challenge to slow adapting credit risk models. From a qualitative point of view, the ability of models to forecast defaults in different stages of the credit cycle is a desirable property, that we will remark in the evaluation of the models' performance.

[8] The status of non-performing credit includes different stages of impairment: past-due 90 days, unlikely to pay and bad loans.

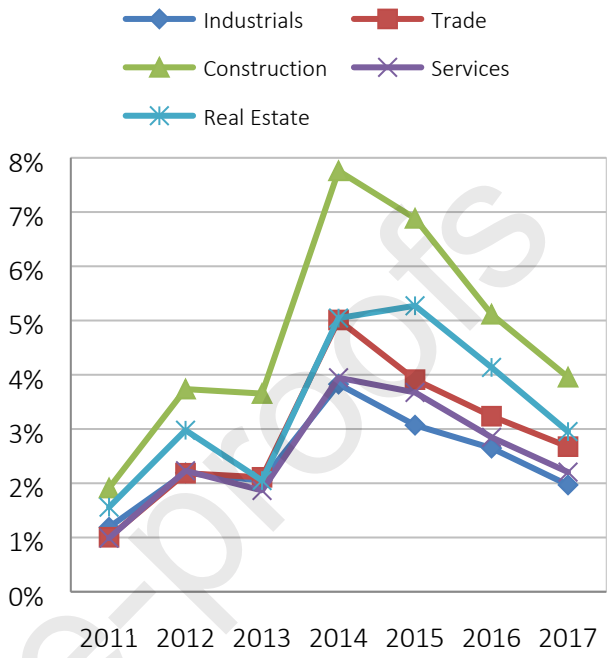
Journal Pre-proofs

Figure 2: Default rate by size and sector

Panel A: firm size



Panel B: economic sector



Note: Our calculation on the Credit Register and Cerved data. Firms' size classes are defined according to the following criteria: micro, less than 10 employees and turnover up to EUR 2 million; small, less than 50 employees and turnover not exceeding EUR 10 million; medium, less than 250 employees and turnover less than EUR 50 million (or total assets not exceeding EUR 43 million); large, all remaining firms. Firm's economic sector affiliation is based on Cerved data.

Structural differences in default risk levels are usually associated with a firm size class and its sector; consistently, credit risk models relate a firm PD to these factors (Altman et al., 2017; Wang and Dwyer, 2011). Larger firms record lower default rates relative to SMEs (in our sample almost 3 percent lower) and these differences magnify in periods of economic turmoil (from 2014 onwards differences in default risk by size class exacerbate). The lower riskiness of larger firms could be explained by their engagement in businesses with a longer history, a larger market share and less volatile revenues.

The economic sector is another relevant dimension of a firm's credit risk, with more cyclical sectors (usually investment goods or real estate) leading to volatile revenues and more fragile firms. Indeed, Construction and Real Estate present significantly higher default rates than the Industrials, with the latter usually including firms characterized by widely diversified businesses with greater exposure to non-domestic markets.

3.2 Financial and credit behavioral indicators

The majority of academic works and industry applications on credit risk modelling uses economic and financial ratios as potential indicators of corporate defaults. We add to those predictors a set of confidential information on the firm-bank relationship embedded in credit behavioral indicators.

Our dataset was drawn from the Company Accounts Data system (provided by Cerved), the National Credit Register (CR) and a confidential dataset maintained by the Bank of Italy (the In-house Credit Assessment System, BI-ICAS[9]), and it contains a wide array of firm-level variables for Italian non-financial companies. Starting from financial ratios we compute twenty-four indicators covering: profitability, financing structure, debt sustainability and asset types. Credit behavioral indicators include eight variables relating to a firm financial flexibility, that is the proportion of drawn to granted bank credit for different facilities, and the occurrence of delinquencies within a firm-bank credit relationship. After including firms' descriptive indicators, such as age, economic sector and graphical area, our set of default predictors contains thirty-eight variables.

Then, variables were selected to be included in the model using the following criteria:

1. using univariate logit regressions for the probability of default, those variables with an accuracy ratio lower than fifty-five percent were dropped;
2. using the Kolmogorov-Smirnov test, those variables with insignificant differences in the distributions between the default and non-default groups were dropped;
3. from the list satisfying 1) and 2), only less correlated variables were retained (rho < 0.7);

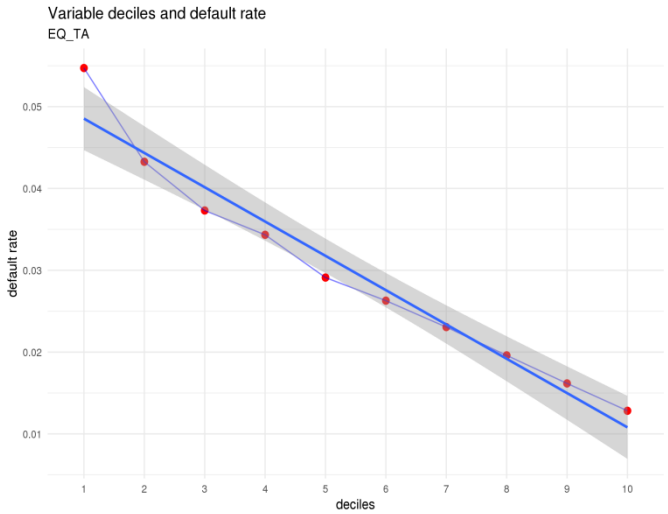
Our final set of default predictors consists of twenty-four variables (see Appendix 1 for a description of the variables), and the estimation dataset includes about 250.000 yearly firm-observations for defaulted and non-defaulted firms.

Those variables displaying a linear relationship with the default rate are suitable candidates as default predictors in all models; for example the financial leverage indicator (equity over total assets) correlates linearly with the default rate (Figure 3, Panel A). Indeed, the most capitalized firms are more resilient to financial distress or economic slowdown. ML models are expected to benefit more than statistical models from non-linear or non-monotonous indicators; for example the sales growth rate indicator displays a non-monotonous relationship with default risk, with low and high sales growth firms being more risky than otherwise (Figure 3, Panel B).

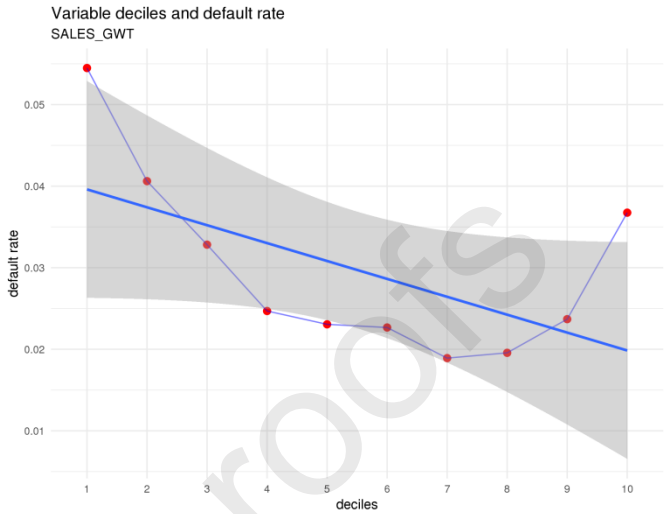
[9] The BI-ICAS is the rating system managed by the Bank of Italy and formally authorized for the valuation of eligible credit claims posted by Italian counterparties as collateral of monetary policy refinancing operation of the Eurosystem. BI-ICAS integrates two credit risk measures, based on a standard logit framework: the 'credit behavior' component, modelling monthly data sourced from the Italian Credit Register; the 'financial' component, based on yearly financial statement data reported in the company accounts data archive of the Bank of Italy. The statistical model of the BI-ICAS produces at a monthly frequency around 300.000 default probabilities for Italian limited companies. The model's performance is controlled every year in the context of the annual monitoring of the accepted credit assessment sources carried out by the Eurosystem.

Figure 3: Default rate and predicting variables deciles

Panel A: Equity over total assets



Panel B: Sales growth



Note: Our calculation on the Credit Register and Cerved data. Figure 3 plots (dots) the default rates associated with different deciles of two of the variables employed to predict defaults.

3.3 Calibration of the models and selection of hyper-parameters

The estimation of credit risk models is subject to the rare-events problem, that is, the dataset usually contains default observations only in a small proportion relative to non-default. In this circumstance, the discriminatory power of a forecasting model is weak: on the one hand, more importance is attributed to those variables that identify sound firms (highly represented in the estimation sample) rather than variables that help discriminate distressed firms; on the other hand, the model tends to give very low default probabilities to all the firms.

Downsampling is commonly used to overcome this issue: in the first stage, the model is estimated on a balanced sample, namely a sample with an equal number of default and non-default observations, with the latter randomly sampled from the original dataset.[10] However, with a default rate equaling fifty percent, the balanced sample does not reflect the real level of credit risk: estimated probabilities of default need to account for the actual default rate of the economy. For this reason, in the second stage, a recalibration is achieved via algebraic manipulation using Bayes correction (see Sugiyama et al., 2017; Dal Pozzolo et al., 2015a).[11]

[10] See Wallace and Dahabreh (2014).

[11] See Appendix 2 for a detailed description of the calibration procedure.

A second issue related to the estimation of ML models is the setting of hyper parameters, which define the general structure of the models. In RDF, we need to calibrate the number of variables selected at each split, whereas in GBT, we need to set the number of trees and leaves of each tree.

Using 5-fold cross-validation, we choose the hyper parameters that maximize the out-of-sample performance of the models.[12] Firstly, we randomly divide the training set into k disjoint subsets (folds) of equal size; then, we train the model k times, each time on a dataset composed by the union of different $k - 1$ folds, and we use the remaining fold to compute an out-of-sample AuROC for the model. Finally, this gives us k out-of-sample accuracy measures for each combination of the parameters, and we select the parameters presenting the highest average AuROC.

4. Results

In this section we compare the credit risk models described in section 2 along their forecasting performance metrics: discriminatory power and precision. First, we evaluate their performance using a limited training dataset which includes financial indicators and firms' characteristics only. This dataset resemble the case of a credit scoring model estimated using only publicly available financial information. Then we use our whole dataset, which covers both publicly and confidential information from the Italian Credit Register; with the latter set of indicator usually available only to lenders or supervisors. Second, we examine the allocation of credit resulting from a credit assignment rule based on the probabilities of default by estimating the amount of credit that each model would allocate (intensive margin), the number of borrowers gaining access to credit (extensive margin), and the default rate a lender would record using a certain credit rating system. Finally, we evaluate the importance that each model gives to individual predictors in terms of their contribution to sort borrowers according to their credit standing. In particular, we compute the decrease of AuROC that a model incurs if a variable is randomly permuted in the dataset before the estimation of the default probabilities.

4.1 Discriminatory power

We assess the discriminatory power of the default probabilities estimated by the different models using the Area Under the Receiver Operating Characteristic Curve (AuROC).[13] This is a measure of the ability of the model to assign higher default probabilities to firms that will default with respect to financially sound firms: a random model that does not discriminate between sound and distressed firms has a 0.5 AuROC, while a perfect model has an AuROC of 1.

[12] For an introduction to cross-validation, see for example Geisser (1993).
[13] See Fawcett (2004), Chawla (2009) and Xu-Ying et al. (2009).

We first report the out-of-sample AuROC for the different models using a restricted set of information to train the model (**Error! Reference source not found.**); this is a dataset limited to financial ratios and firm characteristics (such as geographic area, economic sector and firm size) that can be collected from publicly available sources. The AuROC metric ranges between 72 and 77 percent, a level of accuracy comparable to Wang and Dwyer (2011), Bacham and Zhao (2017), and Barboza et al., (2017).

We find that tree-based models outperform statistical models over the entire time span, with an average increase in discriminatory power over the Logistic (LOG) model of about 2.6 percent. Linear Discriminant Analysis (LDA) and Penalized Logistic Regression (PLR) display results very close to the LOG model, probably due to similarities in their functional forms.

Table 2: Discriminatory power with restricted dataset (financial indicators)

Year	Linear Discriminant Analysis	Logistic Regression	Penalized logistic regression	Random Forest	Gradient Boosted Trees
	LDA	LOG	PLR	RDF	GBT
2012	73,7%	73,9%	73,9%	76,6%	76,3%
2013	73,7%	73,9%	73,9%	77,2%	77,3%
2014	72,2%	72,3%	72,4%	74,4%	73,9%
2015	73,7%	73,7%	73,7%	76,1%	76,0%
2016	72,6%	72,6%	72,6%	75,3%	75,3%
2017	73,0%	73,0%	73,0%	75,7%	75,4%

Note: Our calculation on Cerved data. The AuROC statistic is computed using out-of-sample probabilities of defaults obtained from the various models and observed default data.

We then expand the information set employed to train the different models so as to include credit behavioral indicators. The AuROC for these models are reported in Table 3. We document a noticeable increase in overall discriminatory power when credit behavioral indicators are included: the increase accounts for about 10 percentage points in terms of AuROC. This finding is consistent with Bacham and Zhao (2017), for which credit behavioral indicators lead to similar increase in accuracy over a model based exclusively on financial ratios and firm characteristics. While still outperforming statistical models, tree-based models now provide a smaller increase in discriminatory power over the logistic model. We interpret this finding as the effect of the different information set utilized to train the models: with high quality data, the logistic regression already provides a very

good forecasting performance which approaches the upper bound of feasible values for the discriminatory power; as a result, the improvement in default forecasting due to the use of ML is necessarily less pronounced.

Table 3: Discriminatory power with complete dataset (financial and credit behavioral indicators)

Year	Linear Discriminant Analysis	Logistic Regression	Penalized logistic regression	Random Forest	Gradient Boosted Trees
	LDA	LOG	PLR	RDF	GBT
2012	73,7%	73,9%	73,9%	76,6%	76,3%
2013	73,7%	73,9%	73,9%	77,2%	77,3%
2014	72,2%	72,3%	72,4%	74,4%	73,9%
2015	73,7%	73,7%	73,7%	76,1%	76,0%
2016	72,6%	72,6%	72,6%	75,3%	75,3%
2017	73,0%	73,0%	73,0%	75,7%	75,4%

Note: Our calculation on Cerved and the National Credit Register data. The AuROC statistic is computed using out-of-sample probabilities of defaults obtained from the various models and observed default data.

Finally, we assess the sensitivity of forecasting performance of the different models to the size of the training dataset. To this end, we compile a smaller dataset which includes only 10 per cent of the original observations . Our choice is grounded on the idea that datasets available to individual lenders usually include only a small fraction of the firms covered in our complete dataset. Differently from previous results, we find that tree-based and logistic models achieve substantially similar levels of discriminatory power: ML models need many observations to produce highly accurate forecasts.

Journal Pre-proofs

Table 4: Discriminatory power with a small dataset (financial and credit behavioral indicators)

Year	Linear Discriminant Analysis	Logistic Regression	Penalized logistic regression	Random Forest	Gradient Boosted Trees
	LDA	LOG	PLR	RDF	GBT
2012	82,5%	82,6%	83,5%	83,9%	83,5%
2013	82,8%	82,9%	82,9%	83,0%	83,2%
2014	80,5%	80,8%	80,7%	81,1%	80,8%
2015	82,6%	82,7%	82,7%	83,3%	83,4%
2016	82,6%	82,7%	82,7%	82,7%	82,3%
2017	82,4%	82,6%	82,6%	82,5%	82,2%

Note: Our calculation on Cerved and the National Credit Register data. The AuROC statistic is computed using out-of-sample probabilities of defaults obtained from the various models and observed default data.

In the following sections we concentrate on the default probabilities estimated using the dataset containing only balance sheet indicators and firm characteristics. This is the case where default forecasting can benefit the most from the use of ML models.[14]

4.2 Models discriminatory power across cluster of firms

We decompose the overall gain in discriminatory power attributable to ML models by computing the AuROC of PDs estimated for different clusters of borrowers: sector and size of the firm (Table 5). RDF and GBT increase in discriminatory power appears to be higher for those clusters with lower AuROC for the LOG models, in particular micro and large firms and the construction and agriculture sectors. This result suggests that the relative advantage of using ML models is higher when statistical models report weaker performances; this is probably due to a complex underlying relationship between variables and defaults that statistical models are not able to capture.

[14] Results for the models estimated on the complete dataset containing also credit behavioral indicators from the Italian Credit Register are available upon request.

Table 5: AuROC by sector and size

Sector	Linear Discriminant Analysis	Logistic Regression	Penalized logistic regression	Random Forest	Gradient Boosted Trees
Manufacturing	77,1%	77,2%	77,2%	79,8%	79,7%
Services	72,0%	72,1%	72,1%	74,9%	74,7%
Agriculture	70,8%	71,1%	71,1%	74,9%	73,6%
Energy	70,0%	70,2%	70,2%	72,4%	72,3%
Construction	69,3%	69,4%	69,4%	72,6%	72,2%
Real estate	67,9%	68,0%	68,0%	70,2%	70,3%

Size	Linear Discriminant Analysis	Logistic Regression	Penalized logistic regression	Random Forest	Gradient Boosted Trees
Micro	71,6%	71,7%	71,7%	74,2%	74,1%
Small	77,2%	77,2%	77,3%	79,7%	79,6%
Medium	79,6%	79,6%	79,6%	81,4%	81,3%
Large	74,8%	74,6%	74,7%	77,2%	76,1%

Note: Our calculation on the Credit Register data. Table 5 reports the AuROC statistic computed using out-of-sample probabilities of defaults from the various models, and observed default data. Results are grouped by firms' sector and size.

4.3 Backtesting

To assess how closely probabilities of default match realized defaults we perform a binomial-style test for different credit quality buckets, or steps (CQS). In particular, we use a backtesting strategy outlined in Coppens et al., (2017) as the "traffic light approach". In essence, we test for each bucket how often realized default rates are compatible with the PD forecasts and in the range of usual statistical deviations. A color is assigned based on the p-value of the test, with green indicating that realized defaults are below the expected threshold, and yellow and red indicating that there is a positive or strongly positive discrepancy between expected and realized defaults respectively.

In Table 6 we report the results of the backtesting exercise: each cell indicates the realized default rate within a certain PD bucket, and the PD thresholds represent the upper limit for PD in each interval.

For the years 2013 and 2015-2017, characterized by declining default rates (see Table 1), all of the rating systems report satisfactory performances. Structural models (LDA, LOG and PLR), however, show a lower capacity of correctly classifying high credit quality borrowers, recording several red warnings in the CQSs 1-2 and 3. Tree-based models, on the other hand, do not report significant discrepancies between expected and realized default for these years.

In the years 2012 and 2014, characterized by a strong increase in default risk, the estimated probabilities of default often do not match with realized defaults. Structural models tend to record weaker performances, presenting red warnings in all of the credit quality buckets.

Overall, these results show that the assessment of high credit quality borrowers and the adaptation to rapid deterioration in aggregate default risk is a common problem for structural models, and ML models can (partially) mitigate these issues. In particular, RDF predictions tend to be more precise relatively to the other models in both economic up-turns and down-turns.

Journal Pre-proofs

Table 6: Backtesting

				2012					2013		
CQS	*Treshold*	*LDA*	*LOG*	*PLR*	*RDF*	*GBM*	*LDA*	*LOG*	*PLR*	*RDF*	*GBM*
CQS1-2	*0,1%*	0,4%	0,5%	0,5%	0,0%	0,0%	1,0%	0,6%	0,6%	0,0%	0,0%
CQS3	*0,4%*	0,6%	0,7%	0,7%	0,4%	0,6%	0,4%	0,4%	0,5%	0,2%	0,2%
CQS4	*1%*	1,3%	1,4%	1,4%	1,1%	1,6%	0,6%	0,7%	0,7%	0,5%	0,6%
CQS5	*1,5%*	2,3%	2,5%	2,5%	2,3%	3,1%	0,9%	1,1%	1,0%	0,8%	1,2%
CQS6	*3%*	4,4%	4,5%	4,5%	4,5%	4,9%	1,7%	1,9%	1,8%	1,5%	2,0%
CQS7	*5%*	9,0%	9,0%	9,0%	8,9%	8,1%	3,4%	3,5%	3,5%	3,1%	3,6%
				2014					2015		
CQS	*Treshold*	*LDA*	*LOG*	*PLR*	*RDF*	*GBM*	*LDA*	*LOG*	*PLR*	*RDF*	*GBM*
CQS1-2	*0,1%*	0,5%	0,7%	0,7%	0,0%	0,0%	4,8%	2,7%	2,0%	0,5%	NA
CQS3	*0,4%*	1,0%	1,1%	1,0%	0,2%	0,5%	0,9%	0,8%	0,8%	0,1%	0,1%
CQS4	*1%*	1,4%	1,6%	1,6%	0,7%	1,4%	0,9%	0,9%	0,9%	0,3%	0,5%
CQS5	*1,5%*	2,1%	2,3%	2,3%	1,5%	2,7%	0,9%	1,1%	1,1%	0,6%	1,0%
CQS6	*3%*	3,3%	3,4%	3,3%	3,2%	4,1%	1,7%	1,8%	1,8%	1,4%	1,9%
CQS7	*5%*	5,4%	5,6%	5,6%	6,3%	6,4%	2,8%	2,8%	2,8%	3,1%	3,5%
				2016					2017		
CQS	*Treshold*	*LDA*	*LOG*	*PLR*	*RDF*	*GBM*	*LDA*	*LOG*	*PLR*	*RDF*	*GBM*
CQS1-2	*0,1%*	0,0%	6,4%	7,9%	0,0%	NA	12,5%	2,0%	1,8%	0,4%	0,0%
CQS3	*0,4%*	1,1%	1,0%	1,0%	0,0%	0,1%	0,5%	0,5%	0,4%	0,1%	0,2%
CQS4	*1%*	0,8%	0,8%	0,8%	0,3%	0,6%	0,7%	0,7%	0,7%	0,3%	0,6%
CQS5	*1,5%*	1,0%	1,1%	1,1%	0,8%	1,1%	1,0%	1,0%	1,1%	0,7%	1,2%
CQS6	*3%*	1,8%	1,9%	1,8%	1,6%	2,2%	1,7%	1,7%	1,6%	1,7%	2,2%
CQS7	*5%*	3,6%	3,5%	3,5%	3,5%	3,9%	3,7%	3,7%	3,7%	3,6%	3,7%

Note: Our calculation. The Threshold column reports the upper limit of the CQS interval. For instance, a firm is in CQS3 if its default probability is greater than 0.1 per cent and lower than 0.4 per cent. The percentages represent the realized default rate for each CQS and year.

4.4 Credit allocation and default rates

In this section we evaluate the consequences of using ML rating systems on the allocation of credit. Firstly, we consider the effect on the extensive margin, that is on the number and type of borrowers.

Perspective borrowers are granted access to credit according to their default probability. We compare variations in default rates across the different models; Table 7 shows the results for different amounts of credit, which is set as a proportion (1, 5, 10 and 20) of the total amount of credit drawn in our sample.

While statistical models result in a broader allocation of credit, tree-based model concentrate on a smaller number of safer firms. This pool of borrowers, in turn, would imply a significantly smaller default rate than the statistical models.

Table 7: Credit allocation (fixed granted amount)

% Amount granted	Method	Number of firms	Defaulted amount	Default rate	Default rate - difference wrt LOG
1%	LDA	8.247	29	0,68%	0,12%
	LOG	7.724	23	0,56%	0,00%
	PLR	7.777	24	0,56%	0,00%
	RDF	5.172	10	0,24%	-0,32%
	GBT	5.029	12	0,26%	-0,29%
5%	LDA	19.741	147	0,69%	-0,12%
	LOG	18.713	172	0,81%	0,00%
	PLR	18.951	169	0,79%	-0,01%
	RDF	13.070	106	0,50%	-0,31%
	GBT	13.398	71	0,33%	-0,48%
10%	LDA	29.702	593	1,38%	0,06%
	LOG	28.349	564	1,33%	0,00%
	PLR	28.633	498	1,17%	-0,16%
	RDF	20.022	232	0,54%	-0,79%
	GBT	20.479	349	0,81%	-0,52%
20%	LDA	49.826	1.761	2,03%	-0,03%
	LOG	48.745	1.788	2,06%	0,00%
	PLR	49.300	1.802	2,08%	0,02%
	RDF	34.894	835	0,97%	-1,09%
	GBT	36.458	990	1,14%	-0,92%

Note: Our calculation on the Credit Register data. Allocated amount and defaulted amount in Millions of Euro.

Secondly, we use different thresholds for the default probabilities to decide whether a firm would be granted credit or not:[15]

- PD $\leq$ 0.4% (BBB- investment grade);
- PD $\leq$ 1.5% (BB non-investment grade or high yield);
- PD $\leq$ 5% (B- low grade).

Table 8 shows the amount of credit that would be granted using different rating systems for different PD thresholds. The RDF and GBT models behave rather consistently across the different rating thresholds: both models would grant more credit than the statistical models, with the only exception being GBT in the first PD

[15] The PD thresholds correspond to the Credit Quality Steps (CQSs) of the Eurosystem Credit Assessment Framework (ECAF). Each CQS relates to a rating class utilized by international rating agencies to evaluate the credit quality of a certain issuer.

bucket that allocates roughly the same amount. While extending more credit, this would not result in higher credit losses for lenders: the default rate of RDF and GBT is lower than in other models.

The number of firms which would gain access to credit is always greater with the GBT model than the LOG model. In contrast, the RDF model shows mixed evidence, cutting down the number of firms in the first threshold (with respect to the LOG model), i.e. for the investment grade borrowers, while providing a limited increase for high yield and low grade borrowers.

Overall, the results point to a consistent positive effect of ML on the intensive margin and on the default rate, while the effect on the extensive margin is more complex.

Table 8: Credit allocation (fixed PD thresholds)

Threshold	Method	Number of firms	Default rate	Allocated amount - % difference wrt LOG	Default rate - difference wrt LOG
0,40%	LDA	8.060	0,76%	-46,0%	0,00%
	LOG	12.637	0,75%	0,0%	0,00%
	PLR	12.154	0,74%	-5,6%	-0,02%
	RDF	8.836	0,17%	11,3%	-0,59%
	GBT	15.338	0,28%	113,0%	-0,47%
1,50%	LDA	49.157	1,08%	-9,9%	0,00%
	LOG	53.133	1,12%	0,0%	0,00%
	PLR	52.290	1,10%	-2,8%	-0,01%
	RDF	54.098	0,70%	46,1%	-0,42%
	GBT	74.532	1,02%	104,9%	-0,10%
5,00%	LDA	151.450	2,62%	-0,4%	0,00%
	LOG	151.822	2,62%	0,0%	0,00%
	PLR	151.730	2,62%	-1,1%	0,00%
	RDF	152.906	2,41%	4,5%	-0,21%
	GBT	152.154	2,37%	-0,3%	-0,25%

Note: Our calculation on the Credit Register data. Allocated amounts in Millions of Euro.

Journal Pre-proofs

4.5 Variable importance and model robustness

We analyze the importance that individual predictors have in each model as the extent to which a single variable affects the overall discriminatory power of a model. In particular, for each variable we run the following steps:

1) each firm receives a value for the selected variable that is randomly sampled from the distribution of that variable in the dataset (i.e. we permute the variable across all firms);
2) the AuROC of the model is computed on the permuted dataset;
3) the difference between the AuROC of the model applied to the original test dataset (having all the correct values for the selected variable) and the AuROC obtained on the permuted dataset is our measure of variable importance.

Large values of the difference computed in 3) imply that the variable contributes markedly to the discriminatory power of the model and that, at the same time, the model is very sensitive to potential errors or noise attributable to this variable. Table 9 reports variable importance (AuROC decrease) for each model for the set of explanatory variables. Predictors are ordered according to their average AuROC decrease across all models.

Comparing statistical models and tree-based models we highlight two main results:

- statistical models are more sensitive to the values assumed by single variables: the LOG model, for example, has six variables that induce a decrease in the AuROC of about 3 percent or more when permuted; RDF and GBT have only one. The overall gain in discriminatory power of ML models is, thus, probably attributable to their robustness to noise and to their ability of using the whole information in the dataset.
- variables presenting non-linear and/or non-monotone relationships with default risk, such as Sales Growth, Payables Turnover and Receivables Turnover, are more important for tree-based models than for statistical ones. This piece of result points to a connection between the increase in discriminatory power that ML entail and their capacity to exploit complex relationships between predictors and the outcome variable.

Journal Pre-proofs

Table 9: Variable importance

Variable	LDA	LOG	PLR	RDF	GBT
IE_CASHFLOW	4,5%	3,7%	3,7%	4,5%	5,4%
LOG_ASSETS	3,2%	3,1%	2,9%	1,7%	2,6%
DSCR	1,9%	2,9%	2,9%	2,2%	2,5%
EQ_TA	2,9%	3,0%	3,0%	1,2%	1,6%
TURNOVER	2,9%	2,9%	2,8%	1,4%	1,6%
CASH_ST_DEBT_S	2,4%	2,9%	2,8%	1,0%	2,3%
AREA_CVD	0,9%	0,9%	0,9%	0,3%	0,5%
ATECO_CVD	0,6%	0,6%	0,6%	0,4%	0,3%
PAYABLES_TURNOVER	0,4%	0,3%	0,3%	0,7%	0,6%
EBITDA_MARGIN	0,4%	0,3%	0,3%	0,3%	0,7%
DIM_CVD	0,6%	0,6%	0,5%	0,0%	0,0%
PFN_EBITDA	0,3%	0,3%	0,3%	0,3%	0,4%
SALES_GWT	0,0%	0,0%	0,0%	0,5%	0,9%
VA_TA	0,2%	0,2%	0,2%	0,4%	0,4%
FIN_MISMATCH	0,2%	0,1%	0,1%	0,5%	0,4%
CASH_TA	0,1%	0,2%	0,2%	0,2%	0,3%
RECEIVABLES_TURNOVER	0,1%	0,1%	0,1%	0,4%	0,3%
PFN_PN	0,0%	0,0%	0,0%	0,3%	0,2%

Note: Our calculation on the Credit Register data. See Appendix 1 for a description of the variables. Each column reports the AuROC decrease attributable to a permutation of the variable used in the model.

5. Conclusions

This work compares statistical models usually employed in credit risk modelling with ML models, namely Random Forest and Gradient Boosted Trees. We use a large dataset including financial ratios and credit behavioral indicators for about 300,000 Italian non-financial firms for the years 2011-2017.

When only publicly available information is used to train the models, ML models provide a more accurate forecasting performance, both in terms of discriminatory power and precision, compared to statistical models. This gain is reduced when high quality confidential information, such as credit behavioral indicators, is added to the training dataset. However, when the size of the dataset is insufficient to robustly estimate the relationships

between predictors and the outcome variable, the performance of ML models is similar to that of statistical models.

Moreover, the use of a credit allocation rule based on ML models would result in larger amounts of credit granted and at the same time in lower default rates. This is because ML models tend to concentrate credit towards safer and larger borrowers, resulting in lower credit losses for their lenders.

We argue that the better forecasting performance of ML models is due to their ability of capturing more precisely the complex relationship between the available firms' indicators and the default outcome, demonstrated by the fact that indicators presenting a non-linear or non-monotone relationship with the default are more effectively employed in ML models than in statistical ones. Our results suggest that the joint use of statistical and ML models by lenders or credit analysts may be beneficial for the accurate assessment of potential borrowers. For example, ML models, which are relatively non-transparent, may be used as a benchmark for probability of defaults obtained using more transparent statistical models. This is particularly useful in cases where the estimates derived from the two models are notably diverse, such as larger or riskier borrowers, allowing credit analysts to qualitatively adjust probabilities of default.

Journal Pre-proofs

REFERENCES

Altman, E.I., (1968). Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy. *Journal of Finance*, *23*, 589-609.

Altman, E.I. (1983). *Corporate Financial Distress: A Complete Guide to Predicting, Avoiding, and Dealing with Bankruptcy.* (1st ed.). New York: Wiley.

Altman, E. I., Iwanicz-Drozdowska, M., Laitinen, E. K., & Suvas, A. (2017). Financial Distress Prediction in an International Context: A Review and Empirical Analysis of Altman's Z-Score Model. *Journal of International Financial Management and Accounting*, *28*, 131-171.

Bacham, D., & Zhao, J. (2017). Machine Learning: Challenges, Lessons, and Opportunities in Credit Risk Modeling. *Moody's Analytics Risk Perspectives*, *9*, 30-35.

Barboza, F., Kimura, H. & Altman, E.I. (2017). Machine Learning Models and Bankruptcy Prediction. *Expert Systems with Applications*, *83*, 405-417.

Breiman, L. (2001). Random Forests. *Machine Learning*, *45*, 5-32.

Brown, I., & Mues, C. (2012). An Experimental Comparison of Classification Algorithms for Imbalanced Credit Scoring Data Sets. *Expert Systems with Applications*, *39*, 3446-3453.

Baesens, B., Van Gestel, T., Viaene, S., Stepanova, M., Suykens, J., & Vanthienen, J. (2003). Benchmarking State-of-the-Art Classification Algorithms for Credit Scoring. *Journal of the operational research society*, *54 (6)*, 627-635.

Bhatia, S., Sharma, P., Burman, R., Hazari, S., & Hande, R. (2017). Credit Scoring Using Machine Learning Techniques. *International Journal of Computer Applications*, *161 (11)*, 1-4.

Breiman, L. (2017). *Classification and Regression Trees*. (1st ed.). New York: Routledge.

Chakraborty, C., & Joseph, A. (2017). Machine Learning at Central Banks, *Bank of England Staff Working Paper No. 674*.

Chawla, N. V. (2009). Data Mining for Imbalanced Datasets: An Overview. In *Data mining and knowledge discovery handbook* (pp. 875-886). Boston: Springer.

Coppens, F., González, F., & Winkler, G. (2007). The Performance of Credit Rating Systems in the Assessment of Collateral Used in Eurosystem Monetary Policy Operations. *European Central Bank Occasional Paper No. 65*.

Dal Pozzolo, A., Caelen, O., & Bontempi, G. (2015a). When is Undersampling Effective in Unbalanced Classification Tasks?. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases* (pp. 200-215). Cham: Springer.

Dal Pozzolo, A., Caelen, O., Johnson, R. A., & Bontempi, G. (2015b). Calibrating Probability with Undersampling for Unbalanced Classification. *2015 IEEE Symposium Series on Computational Intelligence*, 159-166.

Fatazzini, D., & Figini, S. (2009). Random Survival Forests Models for SME Credit Risk Measurement. *Methodology and Computing in Applied Probability*, *11*, 29-45.

Fawcett, T. (2004). ROC Graphs: Notes and Practical Considerations for Data Mining Researchers. *Machine learning*, *31 (1)*, 1-38.

Friedman, J.H. (2000). Greedy Function Approximation: A Gradient Boosting Machine. *The Annals of Statistics*, *5*, 1189-1232.

Friedman, J., Hastie, T., & Tibshirani, R. (2001). *The Elements of Statistical Learning*. (1st ed.). New York: Springer.

Fuster, A., Goldsmith-Pinkham, P., Ramadorai, T., & Walther, A. (2018). Predictably Unequal? The Effects of Machine Learning on Credit Markets. SSRN. https://dx.doi.org/10.2139/ssrn.3072038.

Galindo, J., & Tamayo, P. (2000). Credit Risk Assessment Using Statistical and Machine Learning: Basic Methodology and Risk Modeling Applications. *Computational Economics*, *15*, 107-143.

Geisser, S. (2017). *Predictive Inference*. (1st ed.) New York: Routledge.

Khandani, A. E., Kim, A. J., & Lo, A. W. (2010). Consumer Credit-Risk Models via Machine-Learning Algorithms. *Journal of Banking & Finance*, *34*, 2767–2787.

Kruppa, J., Schwarz, A., Arminger, G., & Ziegler, A. (2013). Consumer Credit Risk: Individual Probability Estimates Using Machine Learning. *Expert Systems with Applications*, *40 (13)*, 5125-5131.

Kennedy, K. (2013). Credit Scoring Using Machine Learning. PhD Diss., Dublin Institute of Technology. https://doi.org/10.21427/D7NC7J

Lessmann, S., Baesens, B., Seow, H. V., & Thomas, L. C. (2015). Benchmarking State-Of-The-Art Classification Algorithms for Credit Scoring: An Update of Research. *European Journal of Operational Research*, *247 (1)*, 124-136.

Quionero-Candela, J., Sugiyama, M., Schwaighofer, A., & Lawrence, N. D. (2009). *Dataset Shift in Machine Learning*. (1st ed.). Cambridge: The MIT Press.

Resti, A., & Sironi, A. (2014). *Risk Management and Shareholders' Value in Banking: From Risk Measurement Models to Capital Allocation Policies*. (2nd ed.) New York: Wiley.

Saerens, M., Latinne, P., & Decaestecker, C. (2002). Adjusting the Outputs of a Classifier to New a Priori Probabilities: A Simple Procedure. *Neural computation*, *14 (1),* 21-41.

Swets, J. A. (1988). Measuring the Accuracy of Diagnostic Systems. *Science*, *240 (4857)*, 1285-1293.

Wallace, B. C., & Dahabreh, I. J. (2014). Improving Class Probability Estimates for Imbalanced Data. *Knowledge and Information Systems*, *41 (1)*, 33-52.

Dwyer, D.W., Douglas & Hood, F., Rasch, J. & Salaam, A. (2019). Moody's KMV RiskCalc™ v3.1 Italy.

Liu, X. Y., Wu, J., & Zhou, Z. H. (2009). Exploratory Undersampling for Class-Imbalance Learning *IEEE. Transactions on Systems, Man, and Cybernetics*, Part B (Cybernetics), *39 (2)*, 539-550.

Yobas, Mumine B., Jonathan N. Crook, & Peter Ross. (2000). Credit Scoring Using Neural and Evolutionary Techniques. *IMA Journal of Management Mathematics*, *11 (2)*, 111-125.

Yu, X. (2017). Machine Learning Application in Online Lending Risk Prediction. arXiv:1707.04831. https://arxiv.org/abs/1707.04831.

Yuan, D. (2015). Applications of Machine Learning: Consumer Credit Risk Analysis. PhD Diss., Massachusetts Institute of Technology. http://hdl.handle.net/1721.1/100614.

Zou, H. & Hastie, T. (2005). Regularization and Variable Selection via the Elastic Net. *Journal of the Royal Statistical Society*, *67*, 301-320.

Journal Pre-proofs

APPENDIX 1

We describe the financial and credit behavioral indicators used to predict default of non-financial firms. Variables presenting a non-linear or non-monotonous relationship with respect to the default outcome are labelled with "NL".

Variable	Description
TURNOVER (Asset Turnover Ratio)	Ratio between net sales and total assets. The asset turnover ratio is an efficiency ratio that measures a company's ability to generate sales from its assets.
VA_TA (Value Added to Total Assets)	Ratio between economic value added and total assets. Operating profitability ratio that measures the company's ability to generate value from its assets.
EBITDA_MARGIN (EBITDA to Net Sales) - ***NL***	Operating profitability ratio that measures how much earnings the company is generating before interest, taxes, depreciation, and amortization, as a percentage of revenue.
PFN/PN (Net Debt to Equity)	Measure of a company's financial leverage, calculated by dividing its net liabilities by stockholders' equity.
EQ_TA (Equity to Total Assets)	Ratio between equity and total assets. Used to assess a company's financial leverage
PFN/EBITDA (Net Debt to EBITDA)	Debt sustainability ratio, gives an indication as to how long a company would need to operate at its current level to pay off all its financial debt.
IE_CASHFLOW (Interest Expenses to Cash Flow)	Ratio that indicates the enterprise's ability to pay interest from generated cash flow.
DSCR (Debt Service Coverage Ratio)	Ratio of debt sustainability that refers to the amount of cash flow available to pay interest expenses and annual principal payments on financial debt.
FIN_MISMATCH (financial mismatch)	Mismatch between short term liabilities and short term assets.
CASH_ST_DEBT_S (Current Assets to Short Term Debt)	Liquidity ratio that measures a company's ability to pay off short-term debt obligations with cash and cash equivalents.
CASH_TA (Cash to Total Assets)	Ratio between cash and liquid assets to total assets. It measures a company's liquidity and how easily it can service debt and short-term liabilities if the need arises.
RECEIVABLES_TURNOVER (Receivable Turnover Ratio) - ***NL***	Efficiency ratio that measures how efficiently a company is using its assets. It measures the number of times over a given period (usually a year) that a company collects its average accounts receivable.
PAYABLES_TURNOVER (Payable Turnover Ratio) - ***NL***	Efficiency and liquidity ratio that measures how many times a company pays its creditors over an accounting period.
LOG_ASSETS (Natural Logarithm of Total Assets)	Measures the size of the firm.
SALES_GWT (Net Sales Growth) - ***NL***	Measures a company's growth in a specific year. It also measures the stability of a firm's performance.

Journal Pre-proofs

Variable	Description
DG_CR_TOT (Drawn amount to Granted Amount) - ***NL***	Financial flexibility ratio. It measures the percentage of available credit that the company is actually using. It refers to all the different types of loans.
DG_CR_REV (Drawn Amount to Granted Amount of uncommitted short term loans) - ***NL***	Financial flexibility ratio. It measures the percentage of uncommitted short term loans that the company is actually using.
DG_CR_AUT (Drawn Amount/Granted Amount, short term loans) - ***NL***	Financial flexibility ratio. It measures the percentage of self-liquidating short term loans that the company is actually using.
DUMMY_SCONF (Overdrawns)	Dummy equal to 1 if the firm has an overdrawn amount greater than the granted amount, and 0 otherwise.
DEF_STORIA_CRED (Deteriorated loans)	Dummy equal to 1 if the firm has deteriorated loans, and 0 otherwise.
AREA_CVD (Geographical Area)	Categorical variable identifying the geographical region where the firm operates (North-East, North-West, Center, South and Islands).
MORTGAGE (Mortgage)	Dummy variable equal to 1 if long term loans are more than 90 per cent of total loans. It is used to mitigate the impact on PD estimation of a high drawn/granted ratio which is physiological for mortgages.
DUMMY_REV	Dummy equal to 1 if the firm has uncommitted short term loans, and 0 otherwise.
DUMMY_AUT	Dummy equal to 1 if the firm has short term loans, and 0 otherwise.

Journal Pre-proofs

APPENDIX 2

The performance of credit risk models typically suffers when they are estimated on a dataset that has many more non defaults (healthy firms) than defaults (unhealthy firms). To improve their discriminatory power, a new estimation dataset is derived from the original dataset using a downsampling strategy, consisting of the union of all the unhealthy firms with an equal number of randomly drawn healthy firms.

The in-sample default rate for this new estimation dataset is, by construction, 50 percent, thus the dataset is called balanced; however, while it improves the discriminatory power of the model, downsampling also creates an upward distortion of the estimated probabilities.

Let x be the vector of firm characteristics, D the event that the firm will go into default, and B the event that the firm is selected in the balanced estimation dataset. By training the model on the balanced training dataset we estimate the probabilities $P(D|B, x)$, and not $P(D|x)$, the true probabilities we are actually interested in. However, following a common procedure in the literature, we can link the two probabilities using the Bayes rule:[16]

$$P(D \mid B, x) = \frac{P(D, B, x)}{P(B, x)} = \frac{P(B \mid D, x) \cdot P(D \mid x)}{P(B \mid D, x) \cdot P(D \mid x) + P(B \mid \neg D, x) \cdot P(\neg D \mid x)}$$

$$= \frac{P(D \mid x)}{P(D \mid x) + P(B \mid \neg D, x) \cdot (1 - P(D \mid x))}, \qquad \text{(A.1)}$$

where A.1 follows from the fact that $P(B \mid D, x) = 1$, since all defaulted firms are in the balanced dataset, and that $P(\neg D \mid x) = 1 - P(D \mid x)$. Notice that $P(B \mid \neg D, x)$, the probability that a healthy firm is in the balanced dataset, is known because it is the number of healthy firms in the balanced dataset out of the total number of healthy firms, that by construction is equal to the number of unhealthy firms over the number of healthy firms in the original dataset. If we solve the equation above for $P(D \mid x)$, we obtain:

$$P(D \mid x) = \frac{\beta \cdot P(D \mid B, x)}{\beta \cdot P(D \mid B, x) - P(D \mid B, x) + 1}, \qquad \text{(A.2)}$$

[16] See Saerens et al. (2002) and Dal Pozzolo et al. (2015b)

where we have defined $\beta = P(B \mid \neg D, x) = \frac{\text{\# unhealthy firms}}{\text{\# healthy firms}}$. We use this formula to calibrate the probabilities estimated by the model from the balanced dataset. The calibration procedure is as follows:

1. We estimate the model on the balanced dataset;
2. We apply the model to the (unbalanced) test dataset, obtaining the probabilities $P(D \mid B, x)$;
3. We calibrate the probabilities from 2) using A.2, and obtain the final unbiased probabilities $P(D \mid x)$.

Mirko Moscatelli, Fabio Parlapiano, Simone Narizzano, Gianluca Viggiano**:** Conceptualization, Methodology.

Mirko Moscatelli, Gianluca Viggiano: Software, Data curation,

Mirko Moscatelli, Fabio Parlapiano, Simone Narizzano: Writing- Original draft preparation.

Mirko Moscatelli, Fabio Parlapiano: Writing- Reviewing and Editing.

4.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Journal Pre-proofs

5.

- *Machine learning models entail gains in accuracy relative to statistical models*
- *Due to non-linear relationship between some of the variables and the default outcome*
- *The gains are greater for clusters of borrowers more difficult to predict*
- *The gains decrease when high quality information is used in the training dataset*
- *Credit allocation based on ML models would imply lower default rates*

6.

Journal Pre-proofs